UNIVERSITY OF ASIA PACIFIC

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSE 438: Robotics Lab

BOD- Line Following Car

TEAM MEMBERS:

SHADMAN SARWER (22201242)
H.M. TAHSIN SHEIKH (22201243)
HUMAYRA JIHAN ARPITA (22201244)

SUBMITTED TO:

A S Zaforullah Momtaz (ZAM)
Assistant Professor
Department of Computer Science and Engineering

Project Goals

- Design a high-speed autonomous line-following robot
- Implement a custom 6-channel IR sensor array
- Apply PID calibration control for stable movement
- Improve speed while maintaining tracking accuracy
- Gain practical robotics and embedded systems experience

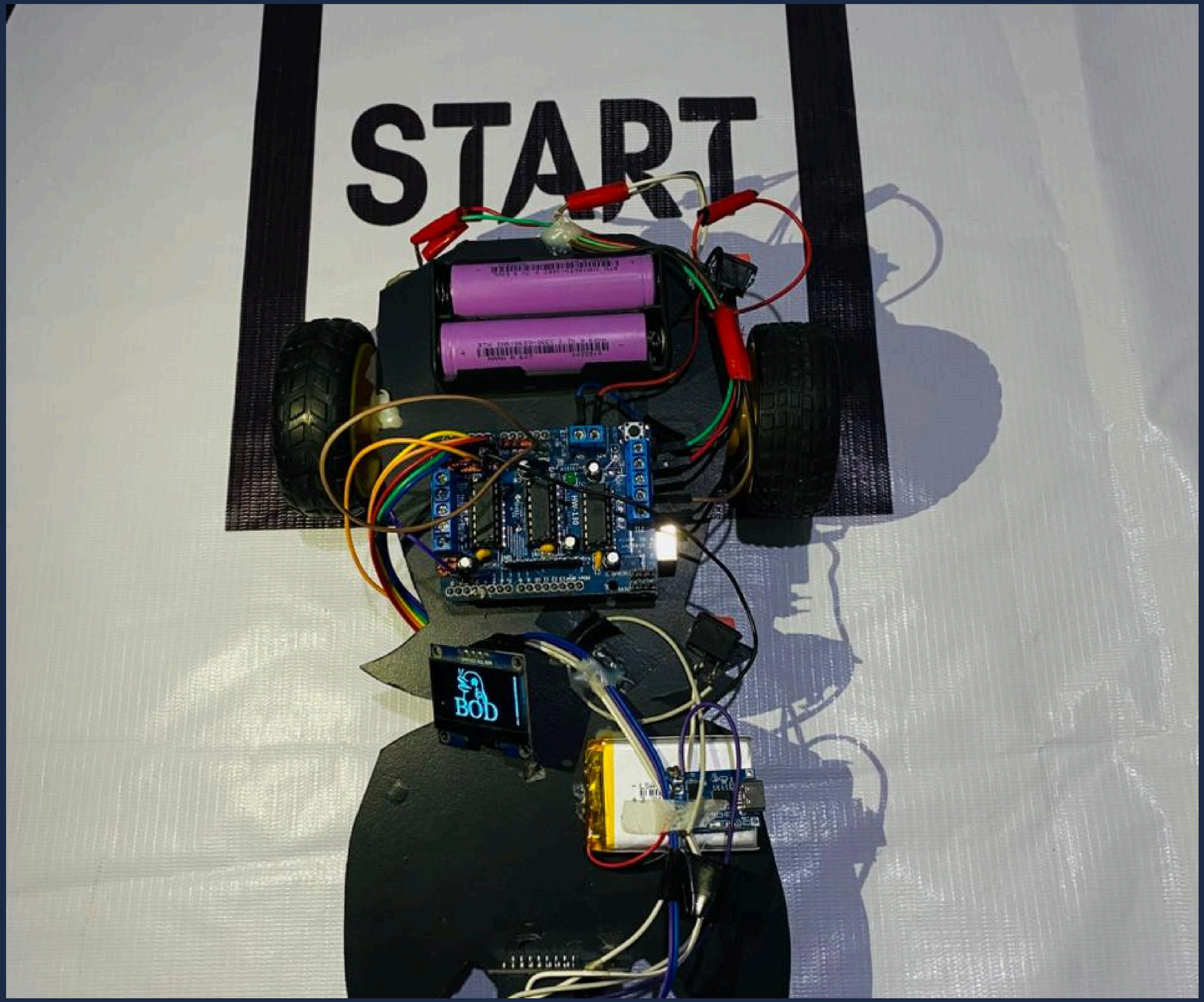


Fig 1: First Overview

Project Objective

- To design and develop an autonomous line following robot.
- To detect and follow a path accurately using a 6-array IR sensor.
- To control robot movement using Arduino Uno and L293D motor driver.
- To achieve fast and stable performance for robot competitions.
- To display sensor readings and system status using an OLED display.
- To improve navigation accuracy on curves and sharp turns.

Social Impact and Motivation

- Reduces human effort through automation and intelligent navigation.
- Demonstrates the practical use of robotics in industries and transportation systems.
- Helps students gain knowledge in embedded systems, sensors, and programming.
- Can be adapted for industrial material transport, warehouse automation, and delivery systems.
- Encourages innovation and interest in robotics and automation technology.
- Motivated by the challenge of building a fast and accurate competition-based autonomous robot.

Implementation / Working Procedure

1. The 6-array IR sensor continuously detects the black line on the track.
2. Sensor data is sent to the Arduino Uno R3 for processing.
3. The Arduino analyzes the sensor readings and determines the robot's direction.
4. The L293D motor driver controls the speed and direction of the two gear motors.
5. The robot moves forward, left, or right based on the detected line position.
6. The OLED display shows team name.
7. Batteries provide power to the robot for stable performance.

Components of BOD Robot

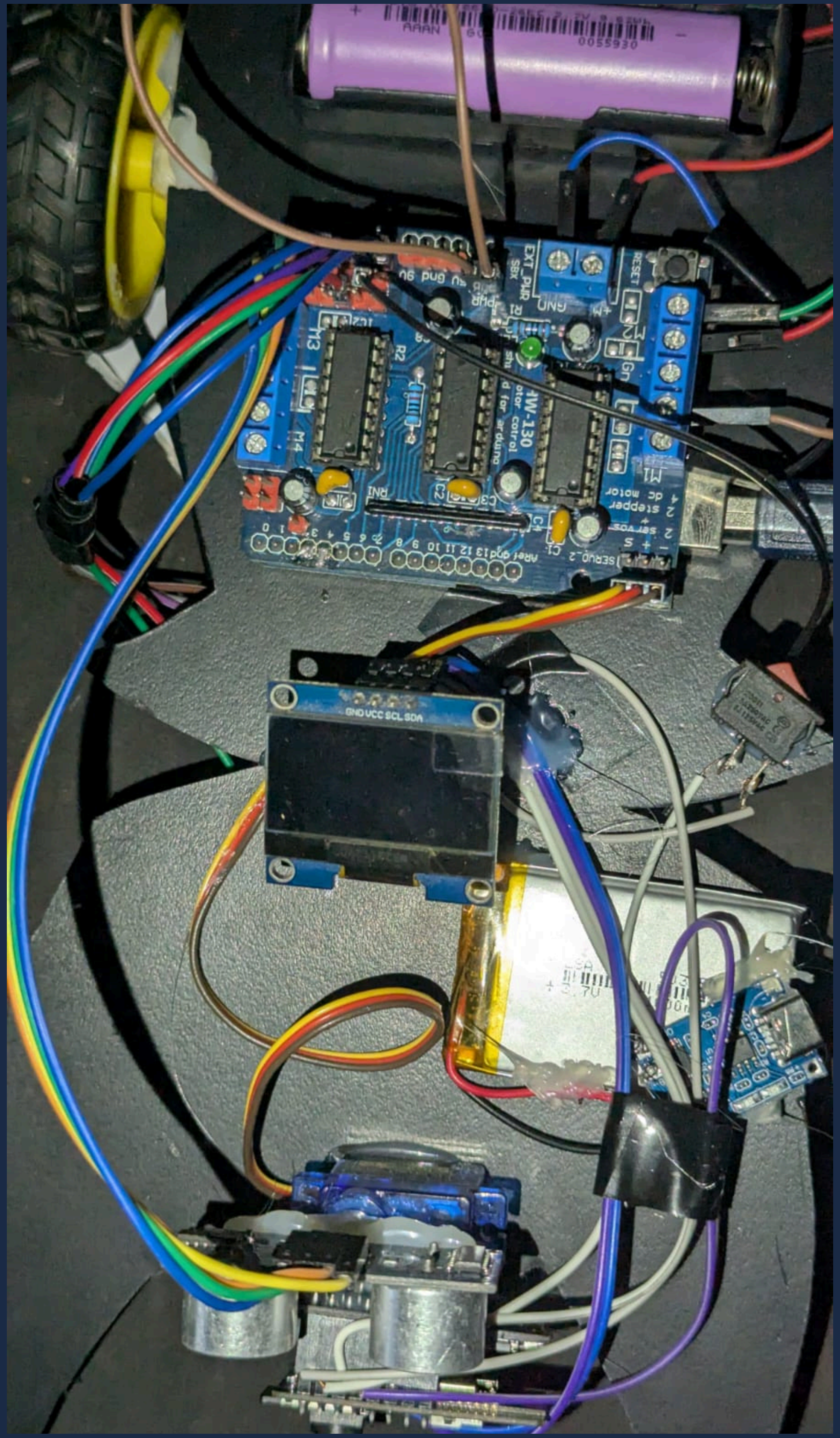
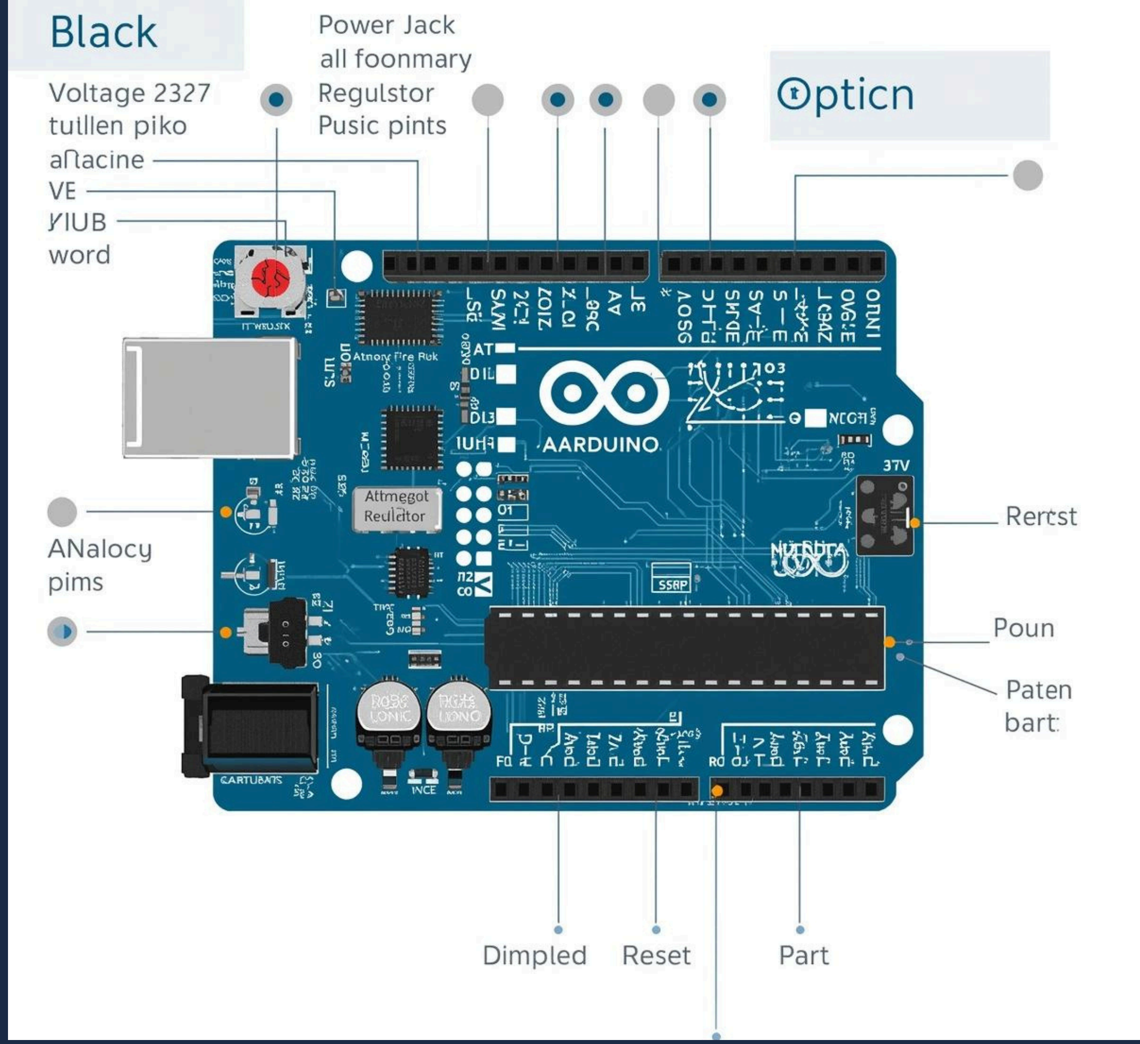


Fig 2: Components

- **Arduino Uno R3** – Main microcontroller for processing sensor and motor data .
- **OLED Display (1.3")** – Shows system status and sensor readings.
- **3.7V 7000mAh Battery** – Powers Arduino, motors, and sensors .
- **3.7V Li-Po Battery** – Powers Display.
- **Gear Motors ×2 with Wheels** – Drive robot forward/backward and turns .
- **N20 Pololu Ball Wheel** – Supports smooth movement and balance.
- **TP4056 5V Charging Module** – Safely recharges batteries .
- **On/Off Switches ×2** – Control power separately for robot and camera .
- **6-Array IR Line Sensor** – Detects line path for navigation .
- **L293D Motor Driver Module** – Controls motor speed and direction.

Components



Arduino Uno

Function:

- Acts as the brain of the robot.
- Reads IR sensors to follow lines and ultrasonic sensor to detect obstacles.
- Controls motors (movement) and servo (sensor scanning).
- Makes autonomous navigation decisions.

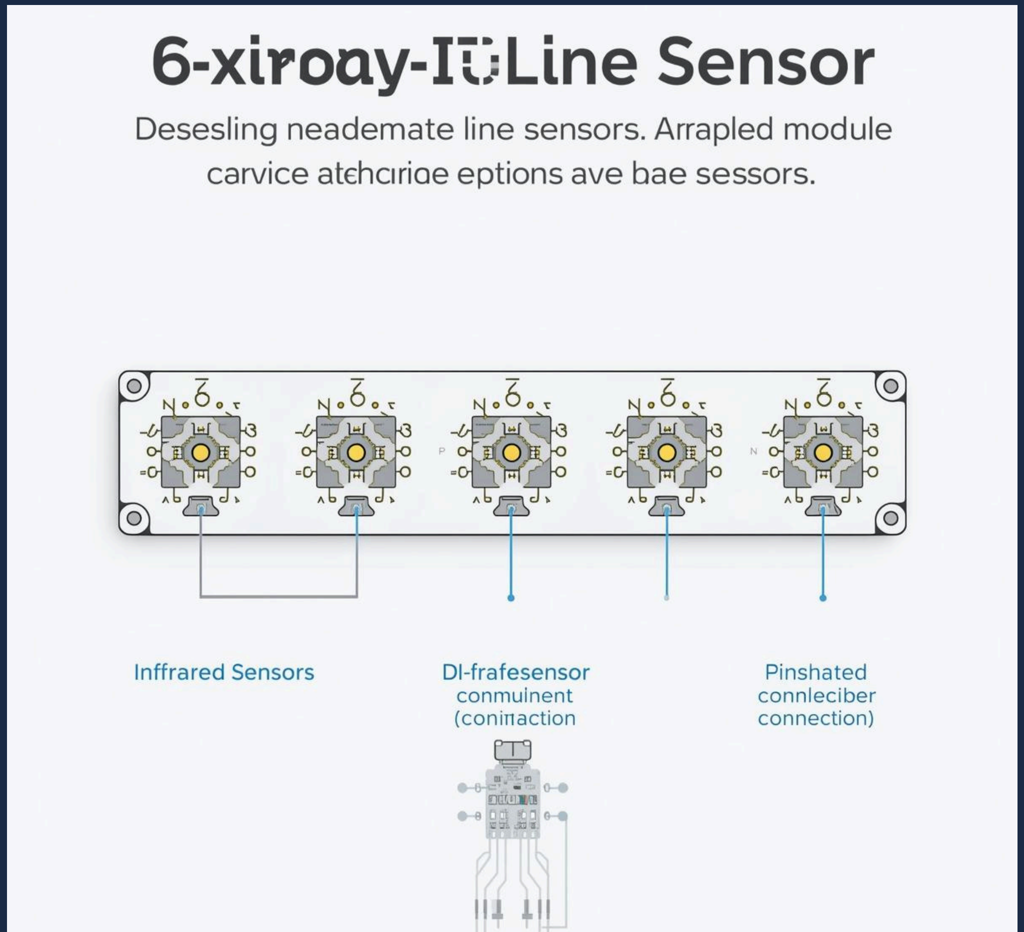
Code Used for Project

```
// Read IR sensors
int d2 = analogRead(ir2);
int d3 = analogRead(ir3);
int d4 = analogRead(ir4);
int d5 = analogRead(ir5);

// Read distance from ultrasonic sensor
int distance = sonar.ping_cm();

// Obstacle detection
if(distance <= 15) {
objectAvoid();  // Avoid obstacle
}
else {
```

```
// Line-following logic
if(d4 < threshold && d3 < threshold) moveForward();
else if(d4 < threshold && d3 >= threshold) moveLeft();
else if(d4 >= threshold && d3 < threshold) moveRight();
else Stop();
}

// Motor control functions
void moveForward() { motor1.run(FORWARD); motor2.run(FORWARD); }
void moveLeft() { motor1.run(BACKWARD); motor2.run(FORWARD); }
void moveRight() { motor1.run(FORWARD); motor2.run(BACKWARD); }
void Stop() { motor1.run(RELEASE); motor2.run(RELEASE); }
```

Components

IR Sensor (6-Array)



Function:

The IR sensor array detects the black line on the surface and helps the robot follow it by sending signals to the Arduino.

Code Used for Project

```
// IR sensor pins
#define ir2 A0
#define ir3 A1
#define ir4 A2
#define ir5 A3
int threshold = 800; // Calibration value

void loop() {
int d2 = (analogRead(ir2) < threshold) ? 0 : 1;
int d3 = (analogRead(ir3) < threshold) ? 0 : 1;
int d4 = (analogRead(ir4) < threshold) ? 0 : 1;
int d5 = (analogRead(ir5) < threshold) ? 0 : 1;
if (d4 == 0 && d3 == 0) {
moveForward();
} else if (d4 == 0 && d3 == 1) {
moveLeft();
} else if (d4 == 1 && d3 == 0) {
moveRight();
} else {
Stop();
}
delay(50);
}
```

System Block Diagram

- This block diagram shows the working of the Smart and fast Line-Following .
- The Arduino Uno controls sensors and motors .
- IR sensors guide the line path.
- L293D motor driver runs the gear motors, and the OLED display shows team name .
- Power is managed by 3.7V batteries and a TP4056 charging module.

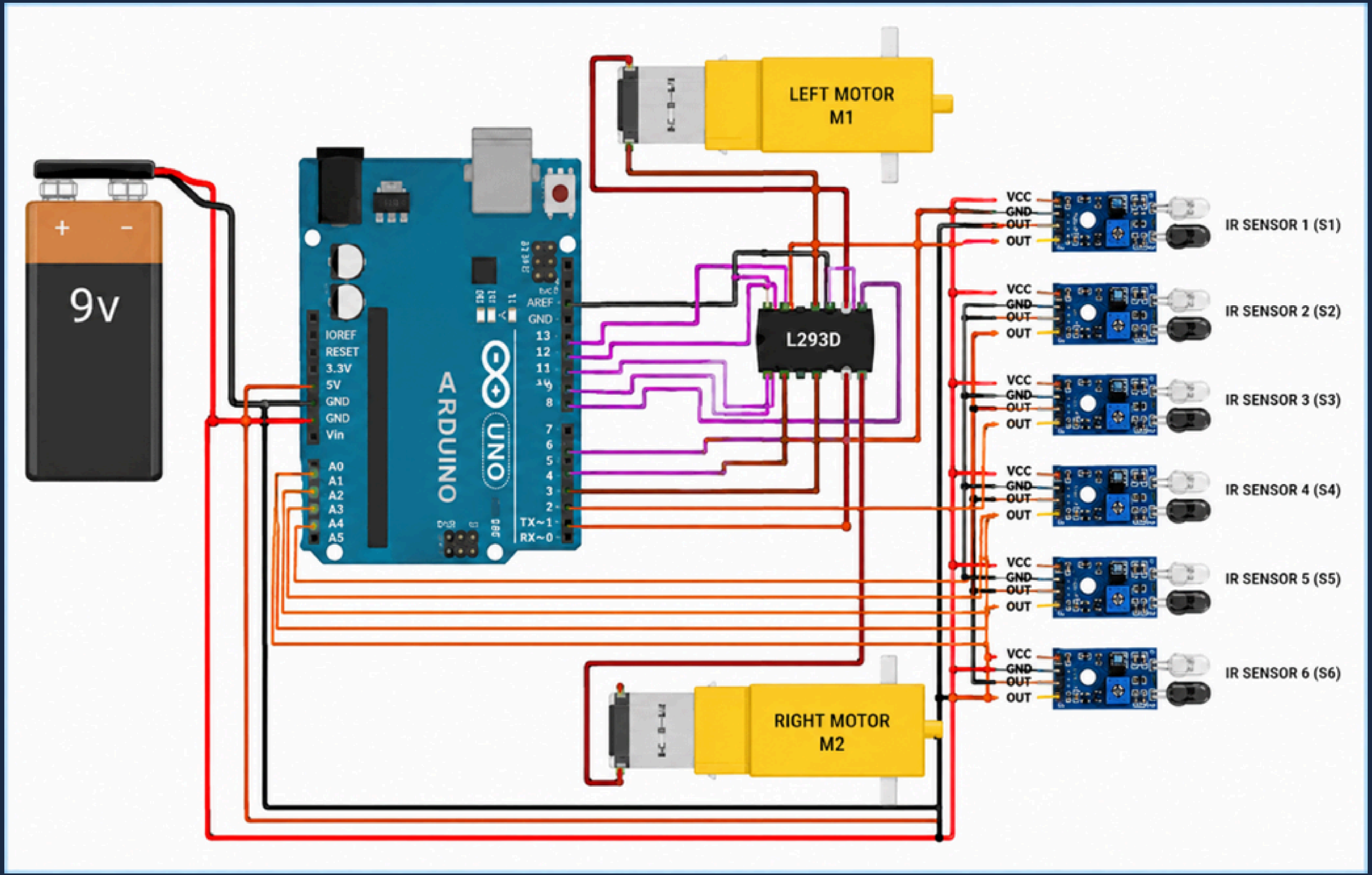


Fig 3 : B.O.D system diagram.

Code Used for Project

```
// Line Following Car - 6 IR Sensor Array - 2WD
// Motor1 = Left wheel (port 1)
// Motor2 = Right wheel (port 2)

#include <AFMotor.h>

// 6 IR sensor pins (S1 = leftmost, S6 = rightmost)
#define S1 A0
#define S2 A1
#define S3 A2
#define S4 A3
#define S5 A4
#define S6 A5

#define THRESHOLD   350
#define BASE_SPEED  140
#define TURN_SPEED  110
#define SHARP_SPEED  70
// 2WD: only two motors
AF_DCMotor motorL(1, MOTOR12_1KHZ);
AF_DCMotor motorR(2, MOTOR12_1KHZ);
bool sensorRead(uint8_t pin) {
 return analogRead(pin) <= THRESHOLD;

void setMotors(int leftSpeed, uint8_t leftDir,
 int rightSpeed, uint8_t rightDir) {
 motorL.setSpeed(leftSpeed);
 motorL.run(leftDir);
 motorR.setSpeed(rightSpeed);
 motorR.run(rightDir);
}
```

```
void goForward()  { setMotors(BASE_SPEED,  FORWARD,  BASE_SPEED,  FORWARD);  }
void stopCar()    { motorL.run(RELEASE);   motorR.run(RELEASE);              }

// Gentle curve: slow down the inner wheel
void curveLeft()  { setMotors(TURN_SPEED,  FORWARD, BASE_SPEED,  FORWARD); }
void curveRight() { setMotors(BASE_SPEED,  FORWARD, TURN_SPEED,  FORWARD); }
id sharpLeft() { setMotors(SHARP_SPEED, BACKWARD, BASE_SPEED, FORWARD); }
void sharpRight() { setMotors(BASE_SPEED, FORWARD, SHARP_SPEED, BACKWARD); }

void setup() {
 pinMode(S1, INPUT); pinMode(S2, INPUT);
 pinMode(S3, INPUT); pinMode(S4, INPUT);
 pinMode(S5, INPUT); pinMode(S6, INPUT);
 Serial.begin(9600);
}
void loop() {
 bool s1 = sensorRead(S1), s2 = sensorRead(S2), s3 = sensorRead(S3);
 bool s4 = sensorRead(S4), s5 = sensorRead(S5), s6 = sensorRead(S6);

 // Debug
 Serial.print(s1); Serial.print(s2); Serial.print(s3);
 Serial.print(s4); Serial.print(s5); Serial.println(s6);

 // Weighted position: negative = line is left, positive = line is right
 int weights[6] = {-3, -2, -1, 1, 2, 3};
 bool sensors[6] = { s1, s2, s3, s4, s5, s6};
```

```
 int activeCount = 0, weightSum = 0;
 for (int i = 0; i < 6; i++) {
   if (sensors[i]) { activeCount++; weightSum += weights[i]; }
 }

 // No sensor: lost line → stop
 if (activeCount == 0) { stopCar(); return; }

 // All sensors: junction or end marker → stop
 if (activeCount == 6) { stopCar(); return; }

 float position = (float)weightSum / activeCount;

 if      (position <= -2.5) sharpLeft();
 else if (position <= -0.5) curveLeft();
 else if (position >=  2.5) sharpRight();
 else if (position >=  0.5) curveRight();
 else                       goForward();
}
```

PROJECT IMPLEMENTATION IMAGE

Prototype Setup:

The prototype of the B.O.D. was built on a two-wheel acrylic chassis powered by Li-ion batteries. The Arduino Uno served as the central controller, interfacing with 6 array IR sensors for line tracking and the L293D motor driver controlled the DC motors.

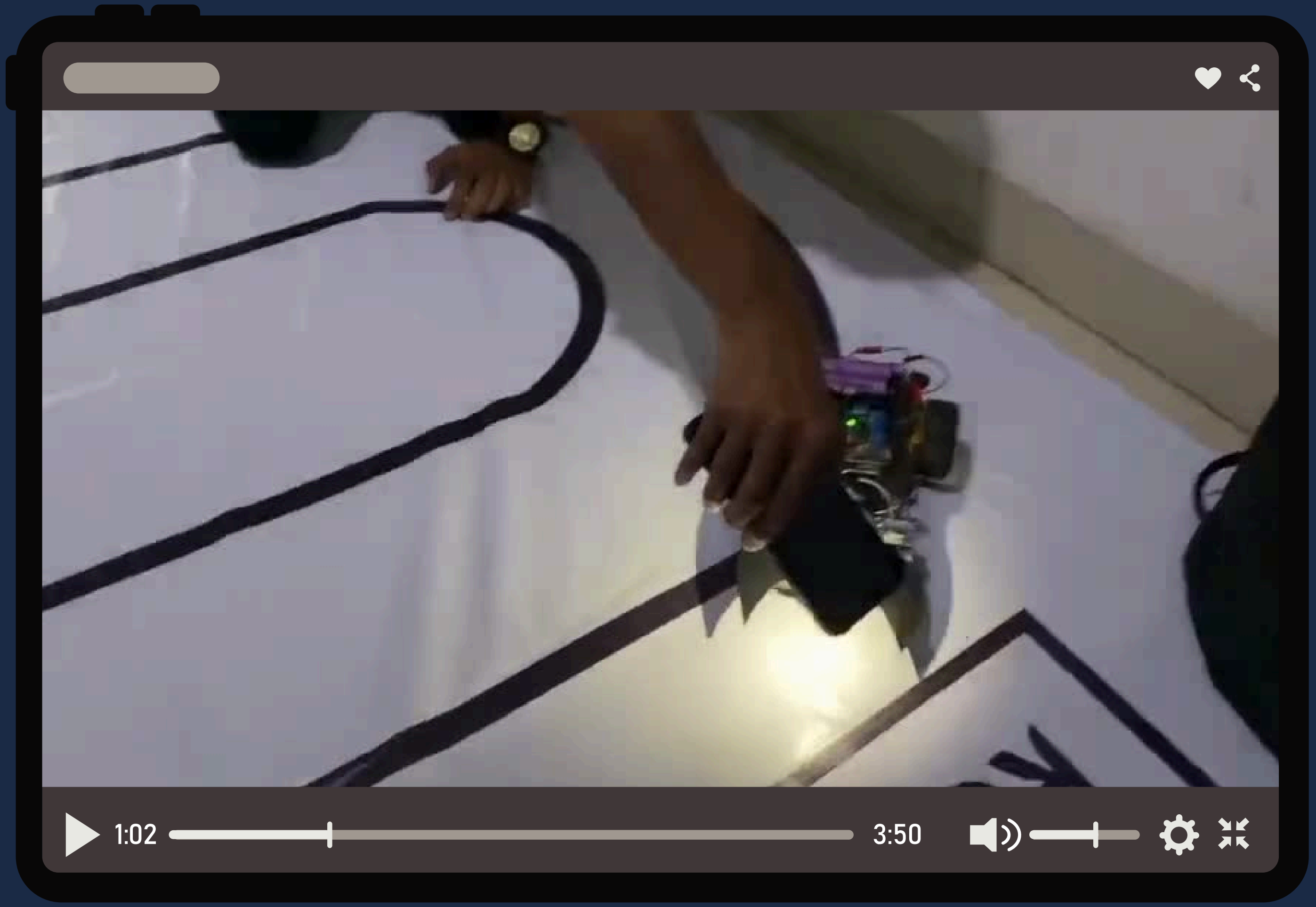
Fig 4: Implementation

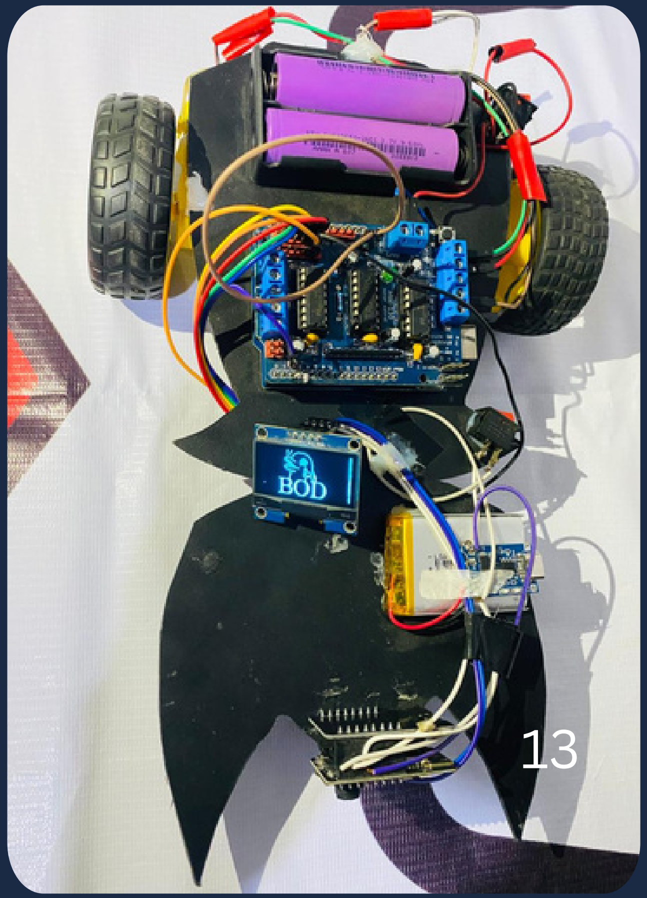
Results and Analysis Slide

Result Summary:

- The robot successfully followed the predefined line path automatically.
- The 6-array IR sensor provided accurate and stable line detection.
- Smooth turning and fast response were achieved on curves and sharp turns.
- The L293D motor driver controlled motor movement effectively.
- OLED display successfully showed real-time sensor and system information.
- Separate power supplies improved overall stability and performance during operation.

OUR PROJECT IN COMPETITION





Conclusion

The project successfully developed a competition-based line following robot using Arduino Uno and a 6-array IR sensor. The robot was able to detect and follow the path accurately with stable and smooth movement. The system demonstrated effective sensor processing, motor control, and autonomous navigation. This project enhanced our practical knowledge of robotics, embedded systems, and automation technology, and it can be further improved for advanced industrial and smart robotic applications.

Thank You

Any Questiom?